# AEGIS: A Resilience-Aware Logistics Planning Engine

Adversarial Disruption Curricula for Classical Planning under Uncertainty

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## 1. Problem Summary

Logistics planning is solved in practice as deterministic cost minimisation. Given a transport network and a set of shipments, a solver (in industry, usually Google OR-Tools) returns the cheapest feasible assignment of shipments to routes and vehicles. The plan is optimal with respect to the cost model it was handed. That model is a snapshot. Roads flood, ports strike, checkpoints slow, warehouses lose power. The plan that was cheapest under the snapshot is often not the plan that degrades most gracefully once the snapshot turns out to be wrong, and nothing in the standard pipeline asks the second question.

This matters because optimisation removes slack. A cost-minimal solution concentrates flow onto the cheapest edges, leaves no spare capacity, and pushes many shipments through a few shared bottlenecks. Those are the conditions under which one localised failure cascades. A slightly more expensive plan that spreads load across two corridors can be worse on the nominal objective and better on the cost actually incurred.

Operations research has a sound formal answer. Robust optimisation minimises worst-case cost or regret over an uncertainty set, and stochastic and dynamic vehicle routing formulations sample scenarios to optimise expected performance. Both are mature, and both depend on the modeller supplying the uncertainty set or the sampling distribution in advance. Plan quality is bounded by what was supplied. A disruption outside the assumed set receives no protection, and neither does one inside it that sits in a tail random sampling rarely draws. Scenario selection is the load-bearing assumption, and the least automated part of the pipeline.

Machine learning has an answer to that problem, developed in an unrelated setting. Domain randomisation trains policies over widely varied simulators, and unsupervised environment design trains an adversary to generate the environments. The central finding is that the right adversarial objective is regret rather than raw difficulty. A pure minimax adversary produces unsolvable and therefore uninformative environments, while a regret-maximising adversary produces environments that are hard but survivable. That result is confined to neural agents trained by gradient descent. A classical planner does not learn from a curriculum, but it can be evaluated against one, and a regret-maximising search over disruption parameters is plausibly a better use of a fixed stress-testing budget than uniform sampling.

A third gap sits outside both fields. Logistics feasibility is not only cost and capacity. Hazardous goods cannot traverse certain tunnels, temperature-controlled cargo has dwell-time limits, cross-

border movements carry documentation requirements. A cost optimiser working from a naive edge-cost model will quietly produce plans that violate these rules, and a resilience evaluation that scores only cost will record a regulatorily infeasible reroute as a successful recovery. Compliance belongs inside the resilience metric.

AEGIS is proposed as a planning layer covering all three gaps. It makes no attempt to out-optimise OR-Tools on nominal cost, since that is a mature CP-SAT implementation. AEGIS generates candidate plans using classical search and planning, annotates the network with disruption probabilities from a Bayesian risk model, checks regulatory feasibility by forward chaining over a goods-and-regulations ontology, then stress-tests each candidate against an adversary that searches for the disruption scenarios inducing greatest regret. Exact minimax with alpha-beta pruning runs below a measured tractability threshold, evolutionary search above it. Once a disruption is realised, the system replans incrementally by patching the affected subgraph.

The deliverable is a measurement rather than a claim. AEGIS and OR-Tools run on identical instances and face identical injected disruptions, and the output is a cost-versus-regret Pareto frontier with both on the same axes, alongside the cost premium AEGIS pays. The expected result is comparable cost with lower regret under disruption. A null result would indicate that cost-optimal plans in this domain are already close to resilience-optimal, and is treated as a real possible outcome.

The research question is narrow and answerable. Does adversarially generated disruption stress-testing expose brittleness in symbolic logistics plans that random or historical scenario sampling misses, at equal compute budget, and what shape does the resulting cost-versus-resilience trade-off take?

## 2. Discussion on related works

(Review of research papers/Patents)

Twenty works are reviewed. Entries 1 to 14 are research papers and the industrial baseline solver, grouped by strand: robust and stochastic optimisation (1 to 6), adversarial curriculum generation (7 to 8), classical planning and replanning (9 to 10), learned search heuristics (11 to 12), and risk modelling with the baseline solver (13 to 14). Entries 15 to 20 are granted patents and published applications, included because the deployed-system view of disruption handling is not visible in the academic literature.

| S. No. | Paper / Patent (Year, Author) | Summary | Methodology | Observations | Gaps / Limitations |
|---|---|---|---|---|---|
| 1 | Aissi, Bazgan & Vanderpooten (2009). “Min–max and min–max regret versions of combinatorial optimization problems: A survey.” European Journal of Operational Research 197(2), 427–438. | Survey establishing min–max and min–max regret as the standard robustness criteria for combinatorial problems under interval and discrete-scenario uncertainty. | Complexity and approximability analysis of the robust counterparts of shortest path, spanning tree, assignment, min cut and knapsack. Exact algorithms surveyed. | Regret against an oracle is a principled robustness objective rather than an ad-hoc metric. Most min–max regret variants are NP-hard even where the nominal problem is polynomial. | Theoretical. The scenario set is assumed given as input, and the survey does not address how to construct or search for the scenarios that matter. |
| 2 | Bertsimas & Sim (2004). “The Price of Robustness.” Operations Research 52(1), 35–53. | Introduces the budget-of-uncertainty formulation, which tunes conservatism with a single parameter Γ while keeping the robust | Linear-programming reformulation of the robust counterpart under a budgeted uncertainty set, with probabilistic bounds on constraint violation. | Robustness is buyable at a modest and quantifiable cost premium. This is the direct ancestor of a cost-versus-resilience trade-off curve. | The uncertainty set is fixed a priori by the modeller. It protects against coefficient perturbation, not against structured discrete disruption |

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| | | counterpart in the same problem class. | | | events such as an edge outage or a depot strike. |
| 3 | Zeng & Zhao (2013). "Solving two-stage robust optimization problems using a column-and-constraint generation method." Operations Research Letters 41(5), 457–461. | The closest existing analogue of this project's adversary loop. A master problem proposes a first-stage decision, an adversarial subproblem generates the worst-case realisation, and the two iterate to convergence. | Master/subproblem decomposition. The max–min adversarial subproblem is reformulated to a single-level MILP via KKT conditions, and its scenario is added back as a new column-constraint pair. | Worst-case scenario generation converges finitely and runs roughly an order of magnitude faster than Benders decomposition on robust location-transportation instances. | The adversary must be expressible as a KKT-reformulable subproblem tied to the LP/MILP structure of the recourse problem, so it cannot wrap a planner whose feasibility depends on ontological or temporal rules. It returns a single robust optimum, never a frontier. |
| 4 | Agra, Christiansen, Figueiredo, Hvattum, Poss & Requejo (2013). "The robust vehicle routing problem with time windows." Computers & Operations Research 40(3), 856–866. | Applies robust optimisation directly to the VRPTW under uncertain travel times. This is the closest OR formulation to the routing half of this project. | Robust MILP formulations (path inequalities, layered-graph and budgeted-uncertainty models) solved exactly by branch-and-cut. | Robust solutions substantially reduce time-window violation under travel-time perturbation for a small increase in nominal cost. | Static: the plan is fixed before uncertainty realises, with no replanning once a disruption occurs. Robustness is reported at a single Γ rather than as a frontier, and no rule-based regulatory feasibility is modelled. |
| 5 | Pillac, Gendreau, Guéret & Medaglia (2013). "A review of dynamic vehicle routing problems." European Journal of Operational Research 225(1), 1–11. | The standard taxonomy of dynamic VRP, classifying problems by how information evolves and how well it is known in advance. | Systematic review with a four-way classification (static/dynamic × deterministic/stochastic) and a survey of solution methods in each quadrant. | Anticipatory policies that exploit distributional knowledge outperform purely reactive re-optimisation in the dynamic and stochastic quadrant. | The dynamism studied is overwhelmingly random customer arrival rather than structured infrastructure disruption. No shared benchmark instances or regret metric exist across the reviewed work. |
| 6 | Ritzinger, Puchinger & Hartl (2016). "A survey on dynamic and stochastic vehicle routing problems." International Journal of Production Research 54(1), 215–231. | Surveys how stochastic information is exploited in dynamic routing, focusing on sampling-based and approximate-dynamic-programming hybrids. | Review of stochastic modelling techniques combined with metaheuristics and ADP/rollout policies. | Scenario sampling from an assumed distribution is the dominant practical anticipation mechanism. | Sampling from the assumed distribution under-represents the tail, meaning the low-probability, high-regret disruptions that resilience planning exists to survive. This is the gap an adversarially generated scenario set is intended to close. |
| 7 | Dennis, Jaques, Vinitsky, Bayen, Russell, Critch & Levine (2020). "Emergent Complexity and Zero-shot Transfer via Unsupervised Environment Design" (PAIRED). NeurIPS 33. | Proposes Unsupervised Environment Design, in which an adversary generates training environments. The key result is that maximising regret rather than raw difficulty produces a usable curriculum. | Three-agent scheme: a protagonist, an allied antagonist, and an environment-generating adversary rewarded by the regret gap between the two students. | Pure minimax adversaries produce unsolvable environments, and domain randomisation cannot adapt difficulty. The regret objective produces environments that are hard but still solvable, and | Assumes a gradient-trained neural student whose learning progress the curriculum shapes. There is no notion of solution cost optimality, no hard feasibility constraint, and no application to logistics or to a non-learning symbolic planner. |

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| | | | | transfers zero-shot to novel levels. | |
| 8 | Tobin, Fong, Ray, Schneider, Zaremba & Abbeel (2017). "Domain randomization for transferring deep neural networks from simulation to the real world." IEEE/RSJ IROS. | The canonical robustness-through-varied-simulation result, and the methodological analogue this project imports into logistics planning. | Randomise simulator parameters (textures, lighting, positions) widely enough that reality appears to the policy as one more sample from the training distribution. | Broad random variation alone yields real-world transfer with no real training data. | Randomisation is difficulty-agnostic, so compute is spent uniformly across easy and hard scenarios. This motivates directing the sampling budget adversarially instead. |
| 9 | Blum & Furst (1997). "Fast planning through planning graph analysis." Artificial Intelligence 90(1–2), 281–300. | GraphPlan, the planning-graph construction and mutex-propagation algorithm underlying this project's symbolic planning core. | Forward expansion of a levelled planning graph with mutual-exclusion propagation, followed by backward solution extraction. | Order-of-magnitude speedups over earlier partial-order planners while remaining sound and complete for STRIPS. | Assumes a deterministic, fully known, static world model. There is no cost model under uncertainty, no probabilistic reasoning, and no replanning once the world diverges from the model. |
| 10 | Fox, Gerevini, Long & Serina (2006). "Plan Stability: Replanning versus Plan Repair." ICAPS, 212–221. | The reference study on whether to patch an existing plan or regenerate it after the execution context changes. This is directly the incremental-replanning question here. | An LPG-based repair strategy modified to favour stability, plus a formal plan-difference metric, compared empirically against replanning from scratch. | Repair produces measurably more stable plans and is frequently faster than replanning, provided the execution context is not highly dynamic. | Evaluated on IPC benchmark domains rather than transport networks with an explicit cost objective. Disruptions are not adversarially selected, and the "highly dynamic" crossover beyond which repair stops paying off is never quantified. |
| 11 | Shen, Trevizan & Thiébaux (2020). "Learning Domain-Independent Planning Heuristics with Hypergraph Networks" (STRIPS-HGN). ICAPS, 574–584. | First method to learn a domain-independent planning heuristic from scratch, establishing that learned heuristics can replace hand-crafted ones. | Hypergraph networks over the delete-relaxation hypergraph, trained on state/cost pairs from optimal plans with a recurrent encode–process–decode architecture. | Competitive with established delete-relaxation heuristics including LM-cut, and generalises to domains unseen during training. | The learned heuristic carries no admissibility guarantee, so A* optimality is forfeited. Per-node inference cost can also offset the reduction in expanded nodes, an effect rarely reported alongside expansion counts. |
| 12 | Chen, Thiébaux & Trevizan (2024). "Learning Domain-Independent Heuristics for Grounded and Lifted Planning" (GOOSE). AAAI / arXiv:2312.11143. | Current state of the art in learned domain-independent heuristics, and the reference point for this project's optional GNN heuristic extension. | GNNs over lifted and grounded graph representations of the planning task, with a formal expressiveness analysis of the resulting heuristics. | Substantially outperforms STRIPS-HGN and generalises to problems far larger than those in the training set. | Carries the same loss of admissibility, and is evaluated on IPC-style domains with no cost-under-risk objective. Whether learned heuristics help on small, dense, weighted transport graphs, where geographic heuristics are already strong, remains untested. |
| 13 | Ojha, Ghadge, Tiwari & Bititci (2018). "Bayesian network modelling for supply chain risk propagation." International Journal of | Models cascading disruption across a multi-echelon supply network with a Bayesian network. This project's risk | Bayesian network over network nodes combined with discrete-event simulation, deriving a risk exposure | Risk propagates measurably. Node-level fragility, service level and lost sales can be predicted under simultaneous | Conditional probability tables are elicited subjectively through expert interviews and are never validated |

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| | Production Research 56(17). | layer follows the same design. | index and a resilience index per node. | disruptions rather than treated independently. | against realised disruption frequencies. The model assesses risk but is not coupled to a planner that acts on it. |
| 14 | Perron & Furnon. Google OR-Tools (CP-SAT solver and routing library), Google, 2011–present. Open-source software. | The industrial baseline this project is benchmarked against, and the de facto standard solver for practical vehicle routing. | Constraint programming with lazy clause generation (CP-SAT) plus routing-specific construction heuristics and guided local search. | Strong cost optimality and maturity on standard routing formulations. A demanding comparison point. | Optimises against a single deterministic cost model known at solve time. There is no built-in stress test, no regret metric, and no way to ask how badly a returned plan degrades if the input model is wrong. |
| 15 | PATENT: US 9,304,006 B2 (granted 2016). “Journey computation with re-planning based on events in a transportation network.” Berlingerio, Botea, Bouillet, Calabrese & Verscheure; International Business Machines Corp. | The closest patented analogue of this project's replanning trigger. It claims a planner that identifies potential future events in advance, monitors for them during execution, and recomputes an improved plan when one fires. | Journey planning over a multi-modal route map. Potential events are enumerated at plan time and stored, the network is monitored by sensors during execution, and detection of an enumerated event triggers recomputation. | The specification explicitly criticises prior systems that replan only once a plan is invalidated, or that brute-force replan at fixed intervals. Anticipatory trigger identification is the right design instinct. | Potential events are enumerated from the existing schedule rather than searched for adversarially, so no worst case is ever sought. The claims are opportunistic, aimed at improving a still-valid plan when something better appears rather than at hardening a plan against loss. No regret metric, and single-traveller journeys rather than multi-shipment fleet plans. |
| 16 | PATENT: US 9,459,108 B2 (granted 2016). “Hedging risk in journey planning.” | Patented treatment of risk-aware rather than purely time- or cost-optimal journey planning, using a hedging framing over uncertain transport outcomes. | Risk-adjusted journey plan computation, selecting plans that trade nominal optimality for reduced exposure to uncertain transit outcomes. | Industry treats the cost-versus-risk trade-off as commercially valuable, and a purely optimal plan is not always the plan a user should be given. | Risk is priced against an assumed distribution over delays rather than probed by adversarial search, so tail scenarios remain under-sampled. There is no compliance dimension and no published benchmark against a cost-optimal baseline. |
| 17 | PATENT: US 10,515,315 B2 (granted 2019; priority IN 201641008563, filed 2016). “System and method for predicting and managing the risks in a supply chain network.” | Machine-learning system for predicting supply chain risk and disruption and recommending mitigation. It is the patented counterpart of this project's risk layer. | Supervised learning over historical supply chain and market signals to score disruption risk across the network, with rule-driven mitigation recommendations surfaced to planners. | Disruption risk scoring is a deployable product capability, and mitigation recommendation is expected to follow prediction. | Prediction and mitigation are decoupled from any optimiser. The system scores risk and advises a human, but never generates or hardens a routing plan against the risk it predicts. No regret metric, and no adversarial probing of the risk model's own blind spots. |

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| 18 | PATENT: US 2014/0046585 A1 (published 2014). "Real-time computation of vehicle service routes." | Fleet-level route recomputation during a disruptive event, and the closest patented match to this project's disruption console and incremental replanning path. | A vehicle management system recalculates scheduled fleet routes on receipt of post-dispatch input, with dedicated flows for dynamic fleet resource management during a disruptive event and for emergency resource management. | Reactive fleet-wide re-optimisation on live disruption input is established, patented commercial practice. | Purely reactive, since the disruption must have occurred before the system acts. There is no pre-deployment stress test, no measure of how much better a differently shaped original plan would have absorbed the same event, and no comparison of patched against full recomputation. |
| 19 | PATENT: US 12,547,793 B2 (IBM; filed 2021). "Digital twin simulation based compliance optimization." Rakshit et al. | The nearest patented relative of this project's compliance-in-the-loop design. A digital twin is simulated forward to predict regulatory violations, and routes are altered in response. | A compliance manager ingests regulatory obligations, builds a digital twin, simulates the consequences of a detected or predicted violation, and generates and executes remediation, including altering a route in light of time of day, road conditions and regulatory requirements. | Coupling regulatory reasoning to simulation and then to routing decisions is a recognised and patentable architecture. This supports placing compliance inside the loop rather than beside it. | The simulation projects consequences forward from an observed or predicted violation. It does not search for the scenarios most likely to cause one. Compliance is enforced but never enters a resilience score, so no cost-versus-regret trade-off is produced. |
| 20 | PATENT (application): US 2025/0173673 A1 (published 2025). "Supply and inventory predictions in supply chain networks." | Recent GNN-based filing indicating where commercial supply chain prediction is heading, and the closest patent to the optional GNN heuristic extension proposed here. | An attention-based graph neural network over supply chain graph structure predicts supply, inventory and imbalance from historical graph data, demand forecasts and original supply plans. Introduces sMACE, a scaled mean absolute cumulative error metric. | Learned graph models materially improve prediction accuracy on real large-scale supply chain data, and the filing concedes that supply executions routinely deviate from initial plans. | Predicts deviation but does not plan against it. The GNN forecasts what will happen, while the question of which plan would best survive that deviation is left unaddressed. It also requires large volumes of proprietary historical data that a synthetic-instance study cannot assume. |

## 3. Outcome on the review:

### Gap/Issue identified and the proposal:

The reviewed work separates into strands that each hold one piece of the problem, and none of which holds all of it.

- **Robust and stochastic optimisation (1 to 6) has the correct objective.** Min–max regret is a well-studied robustness criterion, and two-stage robust optimisation with column-and-constraint generation already implements an adversarial loop. But the adversary in Zeng & Zhao (2013) is a KKT-reformulated MILP subproblem welded to the linear form of the recourse problem. It cannot wrap a planner whose feasibility is decided by symbolic rules over an ontology, and it terminates at a single robust optimum rather than characterising the trade-off curve.

- **Adversarial curriculum generation (7 to 8) has the correct scenario-generation mechanism.** PAIRED establishes that a regret-maximising environment designer beats both uniform randomisation and pure minimax, and that this follows from the objective rather than from any one domain. But it is formulated for gradient-trained neural students, has no concept of solution cost optimality, and has not been applied to logistics or to any non-learning symbolic solver.
- **Classical planning (9 to 12) has the correct solver machinery:** planning graphs, partial-order and conditional planning, plan repair versus replanning, learned heuristics. But it is evaluated on IPC benchmark domains under goal-achievement metrics. Disruptions, where studied at all, are injected arbitrarily rather than adversarially, and no cost-versus-regret frontier is reported.
- **The patent landscape (15 to 20) confirms the gap from the deployment side**, and more sharply than the papers do. Every reviewed filing is reactive or predictive. US 9,304,006 monitors for pre-enumerated events and replans when one fires. US 2014/0046585 recomputes fleet routes once a disruptive event has arrived. US 10,515,315 scores disruption risk and advises a planner. US 2025/0173673 forecasts deviation with a GNN. US 12,547,793 simulates a digital twin forward from a detected violation. None of them searches for the disruption that would hurt the current plan most, then uses that search to select a different plan before anything has gone wrong. US 9,459,108 comes closest by hedging risk at plan time, but prices it against an assumed distribution rather than probing for the tail.

The patent set carries weight beyond citation count. Pre-deployment adversarial stress-testing of a logistics plan is commercially valuable, and every adjacent idea in this space has been filed, so its absence from the set is evidence that the idea is not obvious rather than merely unpublished. These filings treat disruption as something to detect and recover from. None treat it as something to search for deliberately in order to select a more survivable plan.

The gap therefore states in one sentence: no existing work places a regret-maximising adversarial scenario generator around a symbolic, non-learning logistics planner and reports the resulting cost-versus-resilience trade-off against an industrial cost optimiser on shared instances. Each half of that construction is established. The join is not.

AEGIS occupies that join. A classical planning core (BFS/DFS/UCS/A*/greedy search, hill-climbing with restarts, GraphPlan-lite, partial-order and conditional planning) produces candidate plans. A Bayesian risk layer with exact variable-elimination inference annotates edges with disruption probabilities. A forward-chaining compliance engine over a goods-and-regulations ontology decides regulatory feasibility. The adversary module searches the disruption parameter space for regret-maximising scenarios, using exact minimax with alpha-beta pruning below a measured tractability threshold and evolutionary search above it. The meta-controller triggers resilience-weighted regeneration when worst-case regret exceeds a threshold, and incremental replanning patches the affected subgraph when a disruption is realised. All of it is benchmarked head-to-head against OR-Tools on identical instances.

**Clear statement of WHAT'S NEW in the proposal**

Three claims:

- **Solver-agnostic adversarial scenario generation for a symbolic planner:** The adversary treats the planner as a black box scored only by realised regret, rather than as an algebraic subproblem to be dualised. That is what makes it applicable to a planner whose feasibility involves ontological and temporal rules admitting no MILP reformulation, which is the case

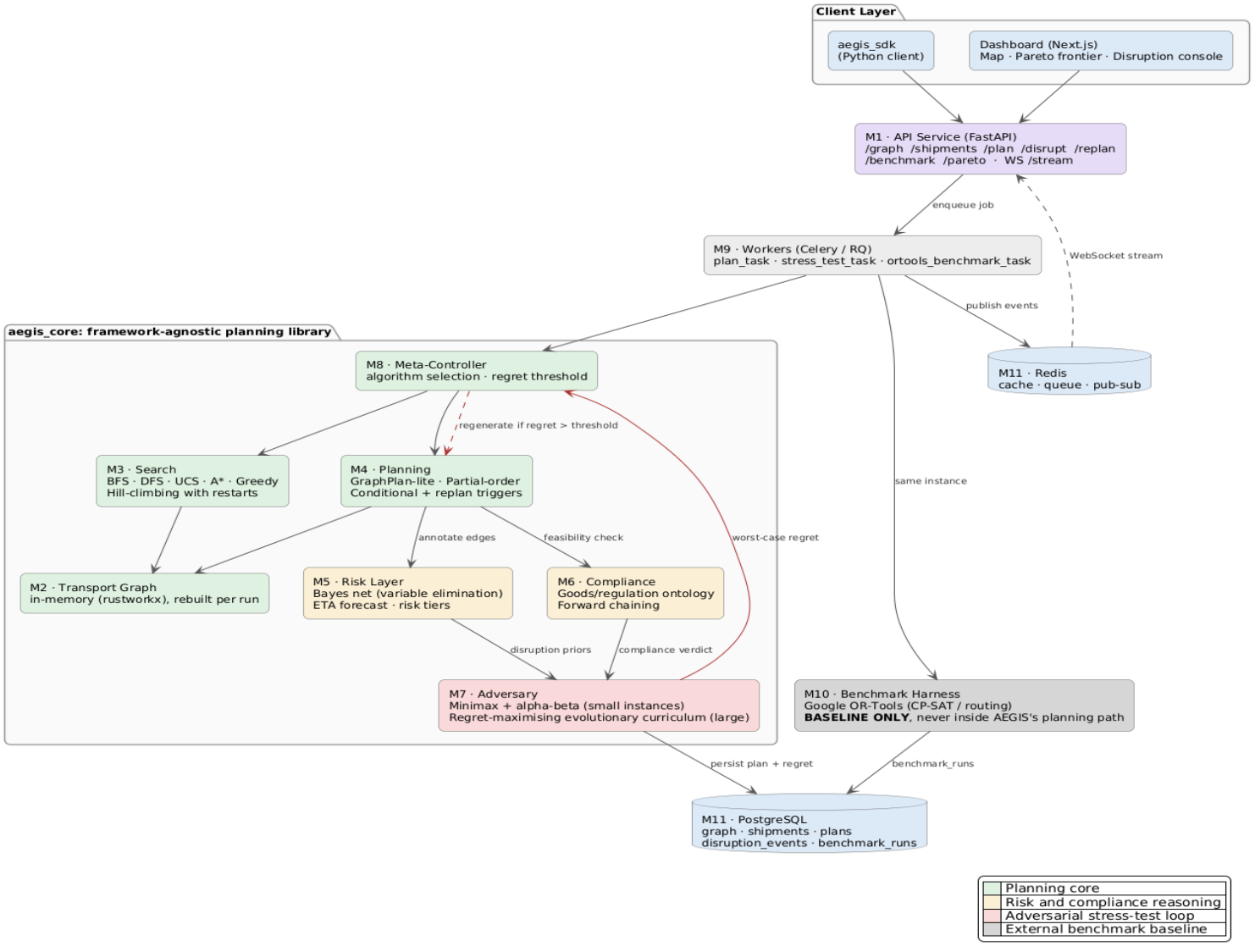
column-and-constraint generation cannot handle. The cost of this generality is the loss of C&CG's finite-convergence guarantee, stated openly rather than glossed over.

- **Transfer of the regret-maximising curriculum objective from RL to classical planning:** PAIRED's finding that regret beats difficulty as an adversarial objective is tested for the first time on a non-learning symbolic solver, where the underlying mechanism differs. There is no student learning curve to shape, only a fixed stress-testing budget to allocate. Whether the finding survives that change is an empirical question this project answers.
- **Compliance-aware regret:** Regret is measured against a compliance-checked oracle, so a post-disruption reroute that is cheap but regulatorily infeasible scores as a failure rather than a recovery. The reviewed robust-routing literature measures cost and time-window violation only, and the reviewed risk work models cascading risk without coupling it to a planner that acts on it.

Explicitly not claimed as new: min–max regret as a robustness criterion, adversarial worst-case scenario generation as a technique, plan repair versus replanning, GNN-learned search heuristics, and beating OR-Tools on nominal cost. Each is established prior work and is cited as such. The novelty claim is confined to the join described above.

## 4. Framework

### Architecture diagram



**Details (Explanation for the modules/blocks one by one)**

**M1, API service.** FastAPI, one router per domain (graph, shipments, plan, disrupt, benchmark, stream). Plan generation is asynchronous and returns 202 with a plan identifier. Full adversarial stress-testing on a realistic multi-depot instance is not a sub-second operation, and a blocking endpoint would misrepresent the cost of the resilience check. A WebSocket endpoint streams plan, disruption and replanning events.

**M2, transport graph.** Depots, routes, per-edge cost, traversal time, goods restrictions and a disruption prior. The graph definition is persisted relationally in PostgreSQL and loaded into an in-memory rustworkx adjacency structure per planning run. No graph database is used, because instances span dozens to a few hundred nodes and are rebuilt per scenario rather than accumulated, so a graph database would add operational overhead for no retrieval benefit.

**M3, search.** BFS, DFS, uniform-cost search, greedy best-first and A*, implemented from scratch. The A* heuristic is great-circle distance scaled by the minimum per-kilometre edge cost, admissible by construction. Hill-climbing with random restarts performs local search over vehicle-to-shipment assignment and stop ordering once a feasible skeleton exists.

**M4, planning.** A GraphPlan-lite planner (levelled planning graph with mutex propagation) for action-level sequencing, partial-order planning for multi-shipment scheduling so that independent shipments are not needlessly serialised, and conditional planning with explicit replanning triggers. Incremental replanning patches only the subgraph affected by a realised disruption, following the plan-repair strategy of Fox et al. (2006).

**M5, risk layer.** A Bayesian network over disruption causes (weather, congestion, industrial action, infrastructure state) and their per-edge effects, with exact inference by variable elimination implemented from scratch. Two scikit-learn models sit alongside it: a regressor for ETA forecasting and a classifier assigning routes to risk tiers. Interpretable models are chosen deliberately, since this module is not the part of the project demonstrating deep learning, and interpretable priors are what make the calibration study in Section 5 possible.

**M6, compliance.** An ontology of goods classes, route attributes and regulations, with forward chaining propagating derived restrictions to fixpoint. It returns a feasibility verdict plus the rule chain that fired, so violations are explainable rather than merely flagged. Its output feeds the regret metric directly, which is what makes regret compliance-aware.

**M7, adversary.** The core research module. Disruption and planner are framed as a two-player game: the adversary picks a disruption vector within a severity budget, the planner responds by replanning, and the adversary's payoff is the induced regret against a compliance-checked oracle. Exact minimax with alpha-beta pruning and iterative deepening runs below a measured instance-size threshold, and beyond it an evolutionary search over disruption parameters (mutating edge targets and severities, selecting on induced regret) takes over. Reinforcement learning is excluded from this phase, because the cheap evolutionary version must first show that a learned curriculum beats random sampling before the expensive version is justified.

**M8, meta-controller.** Selects search and planning strategy by instance size, decides whether the adversary runs exact or evolutionary, and compares worst-case regret against the configured

threshold to choose between finalising a plan and regenerating it under a resilience-weighted cost function.

**M9, workers.** Celery/RQ workers execute the three long-running job types (plan generation, adversarial stress-testing and OR-Tools benchmarking) off the request path, publishing progress events to Redis pub/sub for the WebSocket stream.

**M10, benchmark harness.** Invokes OR-Tools as a library call inside the worker on the same problem instance, then subjects both plans to an identical set of injected disruptions and records the four-way comparison. OR-Tools appears nowhere in AEGIS's own planning path, and this is stated explicitly in the README.

**M11, storage.** PostgreSQL holds depots, routes, shipments, plans, plan stops, disruption events and a first-class benchmark_runs table storing OR-Tools cost, AEGIS cost, and both regret figures per run. Persisting the head-to-head comparison as queryable data rather than script output is what makes the central claim inspectable. Redis provides caching, the job queue and pub/sub.

## 5. Implementation details / Experimental set-up

### Technology stack

Python 3.12 throughout the core. FastAPI (async) for the API, PostgreSQL 16 with SQLAlchemy and Alembic, Redis 7 for cache, queue and pub/sub, Celery or RQ for background jobs, rustworkx for in-memory graph operations, and scikit-learn for ETA regression and risk-tier classification. Google OR-Tools (CP-SAT and routing library) is used as the benchmark baseline only. PyTorch with PyTorch Geometric is reserved for the optional GNN heuristic extension. Frontend in Next.js and TypeScript with deck.gl or Mapbox GL for the route map. Docker Compose across five services (api, worker, frontend, postgres, redis). Observability via structlog, OpenTelemetry and Prometheus, with a run identifier threaded through plan generation, risk annotation, stress-testing and replanning so that a single benchmark run is traceable end to end.

### Instance generation

Synthetic transport networks at three scales: small (10 to 25 depots), medium (50 to 100) and large (200 to 400), across three topology families: hub-and-spoke, grid-like regional, and randomly connected with a tunable density parameter. Edge costs and traversal times are correlated but not identical, since a purely proportional relationship would collapse the trade-off being measured. Shipment sets vary in origin-destination spread, weight distribution, goods class and deadline tightness. Thirty instances per scale and topology cell, with fixed seeds so that every reported result is reproducible from a single command.

### Disruption model

A disruption is a tuple of route identifier, type (weather, industrial action, congestion, infrastructure failure), severity in [0,1] and duration. Severity multiplies edge traversal time and, above a threshold, renders the edge impassable. A disruption scenario is a set of such tuples subject to a global severity budget. That budget prevents the adversary from trivially disabling the network and avoids the unsolvable-environment failure mode Dennis et al. (2020) identify for pure minimax adversaries.

### Baselines and ablations

- OR-Tools CP-SAT / routing solution on the identical instance, the primary external baseline.
- **No-adversary ablation:** AEGIS planning on raw cost with the stress-test loop disabled, isolating the adversary's contribution.
- **Random-scenario ablation:** uniform sampling over the disruption space at equal compute budget to the curriculum generator. This is the central comparison for H2.
- **Historical-distribution sampling:** scenarios drawn from the Bayesian risk layer's priors rather than adversarially.
- **No-compliance ablation:** how frequently a pure-cost plan produces a regulatorily infeasible route.
- Full replanning against incremental patching at increasing disruption frequencies, to locate the crossover Fox et al. (2006) leave unquantified.
- Hand-crafted great-circle heuristic against the optional GNN-learned heuristic, on node expansions, wall-clock time and solution optimality.

**Metrics**

Nominal cost gap against OR-Tools, reported as a signed percentage so that a loss remains visible. Regret against a compliance-checked post-hoc oracle under injected disruption, which is the primary outcome measure. Replanning latency at the 50th, 95th and 99th percentiles. Node expansions and wall-clock time for the heuristic comparison. Compliance violation rate. Adversary efficiency, as induced regret per unit compute against the random-sampling baseline. Calibration error of the Bayesian risk layer against ground-truth injected disruption rates.

**Statistical treatment**

All comparisons are paired at the instance level, since instance difficulty varies far more than the treatment effect being measured. Significance is assessed with the Wilcoxon signed-rank test rather than a paired t-test, because regret distributions are heavily right-skewed by construction. Confidence intervals at 95% come from bootstrap resampling over instances. With thirty instances per cell the study is powered to detect moderate effects only, and the minimum detectable effect size is reported alongside every result. An underpowered comparison is reported as underpowered rather than as a null result.

**Hypotheses under test**

- **H1:** AEGIS plans incur significantly lower regret under injected disruption than OR-Tools' cost-optimal plans, at a small and explicitly stated nominal cost premium.
- **H2:** The adversarial curriculum generator discovers scenarios inducing higher planner regret than random sampling at equal compute budget.
- **H3:** A GNN-learned heuristic reduces A* node expansions relative to the hand-crafted heuristic on large synthetic graphs without materially degrading solution optimality.
- **H0 (null):** No significant regret difference exists between AEGIS and OR-Tools plans under disruption. This is treated as a genuine possible outcome and would be reported as such, indicating that cost-optimal plans in this domain are already close to resilience-optimal.

**Validation and correctness**

Three tiers. Unit tests per module. Correctness tests verify search algorithms against rustworkx and networkx reference output, minimax and alpha-beta against hand-computed values on small game trees with an assertion that pruning never changes the returned value, the planner against a hand-

constructed benchmark set with known-optimal plans, Bayesian inference against brute-force enumeration on small networks, and plan feasibility against OR-Tools' own solution wherever a like-for-like comparison exists. Integration tests run against containerised PostgreSQL and Redis via testcontainers. Property-based tests with Hypothesis assert invariants that must hold for every generated instance: that A* never returns a costlier path than uniform-cost search, and that a compliance-passing plan never contains a restricted goods-route pairing.

**Deliverables**

A reproducible head-to-head benchmark table generated by a single script from freshly generated instances. The cost-versus-regret Pareto frontier with AEGIS candidates and the OR-Tools baseline on shared axes. A regret-over-iterations curve for curriculum-generated against randomly sampled scenarios at matched compute. The measured tractability boundary for exact minimax. A live disruption console demonstrating incremental replanning over WebSocket.